

MODULE 2.1

System and Network Defense

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Prepared by: Kudzaishe Majeza

Physical Security

Topics Covered:

2.1.2 Fencing and Physical Barriers

2.1.3 Biometrics

2.1.4 Badges and Access Logs

2.1.5 Surveillance

2.1.7 Effective Physical Security

2.1.2 Fencing and Physical Barriers

The first line of physical defense — controlling who can enter a site

Purpose of Physical Barriers

- Deter and delay unauthorized access to facilities and sensitive areas
- Create a clearly defined security perimeter
- Work as the outermost layer of defense-in-depth
- Force all entry through controlled access points
- Slow attackers down long enough for detection and response
- Protect against both opportunistic and targeted threats

Types of Physical Barriers

Fencing

Perimeter fencing defines site boundaries and restricts unauthorized entry

Security Guards

Human presence provides real-time monitoring and response capability

Mantraps

Two-door airlocks prevent tailgating — one door must close before the next opens

Turnstiles

Allow one person at a time; prevent multiple people entering on one credential

2.1.3 Biometrics

Using unique physical characteristics to verify identity

What is Biometrics?

- Authentication using unique physical or behavioral characteristics
- Cannot be lost, stolen or forgotten like keys or cards
- Common types: fingerprint, retina scan, hand geometry, voice, facial recognition
- Used for high-security access control
- More reliable than password-based authentication alone
- Often combined with other factors for multi-factor authentication

False Rejection Rate (FRR)

- Also called a Type I Error
- The system fails to recognise a legitimate user
- The authorised person is incorrectly denied access
- High FRR = frustrated users; productivity impact
- Example: fingerprint reader fails to read a worn fingertip
- Reducing FRR may increase FAR

False Acceptance Rate (FAR)

- Also called a Type II Error
- The system incorrectly grants access to an unauthorised person
- A fraudulent user is accepted as legitimate
- High FAR = serious security risk
- Example: retina scanner accepts a similar iris pattern
- The most dangerous error in security contexts

2.1.4 Badges and Access Logs

Smart card access control systems and the audit trail they create

Access Badge Systems

- Smart cards and proximity badges grant physical access to controlled areas
- Each card is linked to a specific user and set of access permissions
- Access rights can be instantly revoked without collecting the physical card
- Badges can be time-restricted — access only during working hours
- Lost or stolen cards are deactivated immediately in the system
- Different zones require different levels of clearance

Card Reader Access Sequence

1 Card Presented

User presents their smart card or badge to the reader

2 Data Transmitted

Reader sends card data securely to the access control server

3 Credentials Checked

Server verifies the card against its access control database

4 Access Granted/Denied

Door unlocks if authorised; alarm triggered if denied

5 Event Logged

Every attempt — success or failure — is recorded with timestamp and location

2.1.5 Surveillance

Monitoring physical spaces to detect, deter and document security events

Why Surveillance Matters

- Deters unauthorized activity — visible cameras reduce opportunistic threats
- Provides real-time monitoring of critical areas and entry points
- Creates a documented record of all physical activity and incidents
- Essential evidence for incident investigation and legal proceedings
- Integrates with access control and alarm systems for automated response
- Covers areas where guards cannot be physically present at all times

Surveillance Methods and Tools

CCTV / IP Cameras

Continuous recording of entrances, server rooms, and sensitive areas

Motion Detectors

Trigger alerts or recording when movement is detected after hours

Security Guards

Live monitoring of camera feeds and physical patrol of the premises

Visitor Logs

Manual or digital log of all visitors including time in, time out, and host

2.1.7 Effective Physical Security

Matching physical security measures to the threats they address

Knowledge Check — Which security measure best addresses each scenario?

Scenario	Best Measure	Reason
Prevent tailgating at entrance	Mantrap / Turnstile	Forces single-person entry; one door must close before the next opens
Verify identity without a card	Biometrics	Uses unique physical traits that cannot be borrowed or stolen
Know who entered which room and when	Access Logs	Every badge swipe creates a timestamped audit record by location
Monitor the server room 24/7	CCTV Surveillance	Continuous video recording captures all activity even without guards present
Slow down an attacker trying to reach servers	Fencing + Guards + Mantrap	Layered physical barriers each add delay, detection and deterrence

Key Takeaways

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Physical barriers — fencing, mantraps and guards — are the first layer of defense-in-depth.



Biometrics use unique physical traits; FRR (Type I) denies legitimate users; FAR (Type II) admits imposters.



Badge access systems control entry to specific zones and every swipe is logged with time and location.



Surveillance through CCTV and monitoring deters threats and creates evidence for investigations.



Effective physical security layers multiple controls — no single measure is sufficient on its own.



Physical security is as important as digital security — most breaches begin with physical access.